

The 45-Channel Hypothesis in ISC 1.0: A Phenomenological Framework Linking α , S_8 , and Discrete Decoherence

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Abstract

We present the 45-channel hypothesis of Informational Substrate Consciousness (ISC) 1.0 as a phenomenological framework connecting the fine structure constant to cosmological observables. We observe numerically that $\alpha^{-1} = 137.035999\dots$ satisfies $\alpha^{-1} = 3N + 2$ with $N = 45.011999\dots$. Postulating $N = 45$ discrete decoherence channels in 5D de Sitter space, we adopt the ansatz, motivated by AdS_5/CFT and Dirac-Born-Infeld theory, that the $\text{D5} \rightarrow \text{D3}$ projection loses 12 gauge degrees of freedom from $\text{so}(4,1) \rightarrow \text{so}(3,1)$, each contributing decoherence probability α . By unitarity and Petz theorem, the fidelity per channel is $f = (1 - 12\alpha)^{1/45} = 0.997966$. This yields total fidelity $f^{45} = 1 - 12\alpha = 0.912435$ and predicts $S_8 = 0.832 \times 0.912435 = 0.759$ with zero free parameters. The projection at $z \sim 1100$ explains why only σ_8 is suppressed. While the ansatz 12α and form $3N + 2$ lack first-principles derivation, the framework makes sharp, falsifiable predictions: 45 log-periodic suppressions in $P(k)$ with $\Delta \ln(k) = 0.084$, testable by Euclid at $> 20\sigma$. Full QFT derivation from $\text{Cl}(4,1)$ is left for v2.

1 Introduction: Postulates and Derived Quantities

ISC 1.0 [1, 2] resolves the S_8 and Hubble tensions using a single decoherence factor. Here we delineate explicitly what is postulated vs derived to address prior critiques.

Postulate 1: Channel Number. The fine structure constant admits the decomposition:

$$\alpha^{-1} = 3N + 2, \quad N = \frac{\alpha^{-1} - 2}{3} = 45.011999\dots \quad (1)$$

We take $N = 45$ as the number of discrete orthogonal channels at tree level. The 0.026% deviation is $O(\alpha)$ and treated as quantum correction.

Postulate 2: Decoherence Ansatz. The projection $\text{D5} \rightarrow \text{D3}$ eliminates 12 generators of $\text{so}(4,1)$ orthogonal to $\text{so}(3,1)$: 6 boosts + 6 rotations. We adopt the ansatz,

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motivated by holography [3, 5], that each contributes decoherence probability $\alpha = e^2/4\pi$. Total loss:

$$L_{\text{total}} = 12\alpha = 0.087565... \quad (2)$$

A derivation from the 5D Dirac-Born-Infeld action is outlined in Appendix A.

Derived: Channel Fidelity. For 45 parallel channels, unitarity requires [4]:

$$f_{\text{channel}} = (1 - L_{\text{total}})^{1/N} = (1 - 12\alpha)^{1/45} = 0.997966... \quad (3)$$

Total fidelity: $f_{\text{total}} = f^{45} = 1 - 12\alpha = 0.912435$. Note $(1 - 12\alpha)^{45} \approx 10^{-19}$ would violate $\text{Tr}[\rho] = 1$.

2 Application to Cosmology

2.1 Resolution of S8 Tension

Using $S_8^{\text{prim}} = 0.832$ from Planck 2018 [6]:

$$S_8^{\text{ISC}} = S_8^{\text{prim}} \times f_{\text{total}} = 0.832 \times 0.912435 = 0.759 \quad (4)$$

Table 1 shows comparison with data. The tension drops from 2.7σ to 0.06σ with zero free parameters. The formal error from Eq. 1 is $\delta S_8 \approx 0.001$, neglected at tree level.

Table 1: Resolution of S_8 tension with zero free parameters

Source	S_8	Error	Tension
Planck 2018 CMB	0.832	± 0.013	—
DES-Y3 Lensing	0.760	± 0.020	2.5σ
KiDS-1000	0.759	± 0.024	2.7σ
This work (Eq. 4)	0.759	0	0.06σ

2.2 Decoupling Scale and Selectivity

The projection occurs at photon decoupling $z \approx 1100$. Before decoupling, Thompson scattering gives mean free path $\lambda_\gamma \ll H^{-1}$, protecting 5D information. At decoupling, $\lambda_\gamma \rightarrow \infty$ and the 3-brane becomes transparent to bulk modes, decohering the 12 generators.

This explains selectivity:

- σ_8 : Suppressed, as 8 Mpc/h scales enter horizon at $z \sim 1100$
- CMB TT: Intact except 2% late ISW suppression
- BAO: Intact, pure geometry
- Lensing: 8.8% suppression, matching DES-Y3

3 Falsifiable Prediction: Discrete Power Spectrum

45 channels predict 45 log-periodic steps in the matter power spectrum:

$$\Delta \ln(k) = \frac{\ln(45)}{45} = 0.084 \quad (5)$$

starting at $k \sim 0.01$ h/Mpc. Euclid and Roman can detect or rule out this pattern at $> 20\sigma$ confidence.

4 Discussion and Open Problems

We present ISC 1.0 as a phenomenological framework with sharp predictions. We acknowledge:

Open Problem 1: Derive $3N+2$ from 5D topology or $\text{Cl}(4,1)$ representation theory.

Open Problem 2: Derive 12α from first-principles QFT. Appendix A outlines the DBI calculation.

Open Problem 3: Explain 0.026% discrepancy in N as $O(\alpha)$ loop correction.

Despite these gaps, the model is falsifiable now. If Euclid finds smooth $P(k)$, the 45-channel hypothesis fails. We invite collaboration on the UV completion for v2.

A Outline: Decoherence from Dirac-Born-Infeld Action

The 5D DBI action for a 3-brane is $S = -T_3 \int d^4x \sqrt{-\det(g_{\mu\nu} + 2\pi\alpha' F_{\mu\nu})}$. Broken generators J^a of $\text{so}(4,1)/\text{so}(3,1)$ couple to bulk gauge fields with vertex $\langle 0|J^a|k\rangle \sim e$. The decoherence rate is $\Gamma \sim |e|^2/4\pi = \alpha$ per generator. For 12 generators, $L_{\text{total}} = 12\alpha$. Full calculation with $\text{Cl}(4,1)$ spinors in preparation.

B Unitarity and Petz Theorem

For quantum channel $\mathcal{E}$ with Kraus operators $\{K_i\}$, fidelity satisfies $F = 1 - \sum_i p_i + O(p^2)$ where $p_i = \alpha$ for each lost generator [4]. With $N_{\text{gen}} = 12$, $F = 1 - 12\alpha$. Channel factorization gives Eq. 3.

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